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Maritime Trade and Conflict: An Empirical Study of Nigerian Regions

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Dedicato a SCRIVI QUI LA DEDICA

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Chapter 1

Introduzione

1.1 Motivation and Index Construction

While a vast literature has explored the effects of trade liberalization in developed economies, less is known about its consequences in developing countries, particularly concerning internal stability and conflict. In these contexts, trade represents a major channel through which global economic fluctuations are transmitted to local welfare and political stability (UNCTAD, 2012). Existing research has often examined related mechanisms through the lens of commodity price shocks. For instance, Dube and Vargas (2013) and Brückner and Ciccone (2010) analyse how international price variations affect conflict outcomes, finding mixed results depending on the type of resource and the institutional environment. Other contributions have instead focused on the reverse relationship, investigating how conflict disrupts trade flows and economic integration (Martin et al., 2008). However, evidence on how trade shocks themselves, rather than price or conflict shocks—affect local stability remains limited, especially at the subnational level in low- and middle-income countries. Our analysis addresses this gap by examining how spatial exposure to trade integration shapes conflict dynamics within Nigeria, combining detailed information on port activity, global connectivity, and local socio-economic characteristics. Our analysis builds on the intuition of Berman and Couttenier (2015), who show that positive income shocks, such as those generated by trade, can reduce conflict by improving local economic opportunities. Conversely, trade can also exacerbate tensions when it increases the value of contestable rents or reinforces existing inequalities, as shown by Amodio et al. (2024). These dual mechanisms, often referred to as the *opportunity* or *income* and *rent-seeking* effects, suggest that the impact of trade on conflict is theoretically ambiguous and context-dependent.

Nigeria represents an ideal setting to study these dynamics. It is a coastal country with a diversified set of ports and abundant natural resources, where trade-related income shocks can have significant spatially heterogeneous effects. Its geography—marked by strong North–South disparities and localized ethnic divisions—makes it a relevant case for testing whether trade integration promotes peace or amplifies existing inequalities.

To empirically assess these mechanisms, we construct a spatial trade exposure index inspired by shift–share approaches. The index combines three main compo-

nents:

1. **Trade shocks**, measured as annual changes in port-level container throughput from IMF PortWatch;

2. **Connectivity**, proxied by the Liner Shipping Connectivity Index (LSCI) from UNCTAD, which captures each port's integration in global maritime networks;

3. **Proximity**, expressed as the inverse distance between each ADM2 centroid and the nearest port (including Onne, added to complement the FAO–UNCTAD list).

INDEX This formulation captures how strongly each subnational unit is exposed to international trade shocks depending on its geographical proximity and the global relevance of its nearest ports. Regions located closer to active and well-connected ports are assumed to experience stronger transmission of trade-related income shocks.

Our contribution is twofold. First, we provide one of the first subnational evaluations of trade shocks and conflict in Africa, combining spatially detailed datasets on trade, conflict, infrastructure, and environmental factors. Second, we show that the effect of trade openness is highly heterogeneous: while it fosters stability in resource-constrained and less developed areas, it may reinforce instability in already prosperous or rent-dependent regions. These findings highlight the importance of considering spatial inequality in designing conflict-sensitive trade and infrastructure policies.

Chapter 2

Data

In our empirical analysis, we combined different data sources to derive a panel of subnational geographical units for the period from 2009 to 2019.
Conflict. Data on conflict event are drawn from the "Armed Conflict and Location Event" ACLED (2025) which cover conflict event from 1999 to 2024. The dataset provides the exact location of events, the disorder type, and other characteristics related to the conflict. (Raleigh et al., 2010)
In constructing the dataset, we focused on consumption-related conflict events rather than rapacity-based ones. Both theoretical and empirical considerations guide this choice: standard trade models usually predict that trade openness influences a country's income, usually in a positive way. However, globalisation has not always reduced inequalities in emerging countries, causing unequal distribution. (Dix-Carneiro & Kovak, 2023) As stated by Dube and Vargas (2013), an increase in income can reduce conflict, but on the other hand, higher income means there is more to fight over these resources. The existence of two opposing influences, opportunity cost and rapacity, suggests that some shocks may mitigate conflict, meanwhile other shocks could exacerbate it, depending on the relative strength of the two effects. The study of Amodio et al. (2024) shows that exposure to exports increasess simultaneously economic activity and political violence and McGuirk and Burke (2020) have shown that, if there is a positive price shocks, producers' revenues increase but consumers' real incomes decrease, enhancing the probability of protests or demonstrations. Conversely, Berman and Couttenier (2015) find that at the micro level positive income external shock, such us those generated by the trade, tend to have a pacifyng effect on conflict. In general, consumption-related mechanisms are more directly influenced by exogenous trade shocks, while conflicts over resources follow different dynamics, often linked to the strategies of armed groups. From an empirical point of view, limiting to consumption-related events also mitigates concerns about endogeneity: resource predation can in itself influence local trade flows, risking to reflect both cause and effect simultaneously.
The final variable therefore measures the annual number of consumption-related conflict events for ADM2, filtered to retain only: (i) subcategories of events not related to predation, such as protests, demonstrations, arrests, agreements or territorial changes not directly linked to resource predation. (ii) Events reported in narrative notes as disputes over consumer goods or livelihoods (keywords such as land, livestock,

pasture, farm, grazing, oil, gold, looting). Events clearly associated with predatory motives (e.g., looting, armed clashes over resource extraction, kidnappings for ransom) are excluded.
After filtering, we limited the dataset to events occurring in coastal regions, then projected them into a consistent spatial reference system and overlaid them on Nigeria's second-level administrative units (ADM2). For each ADM2-year, the count of non-predatory conflict events is available, which serves as the outcome variable in my regression analysis.

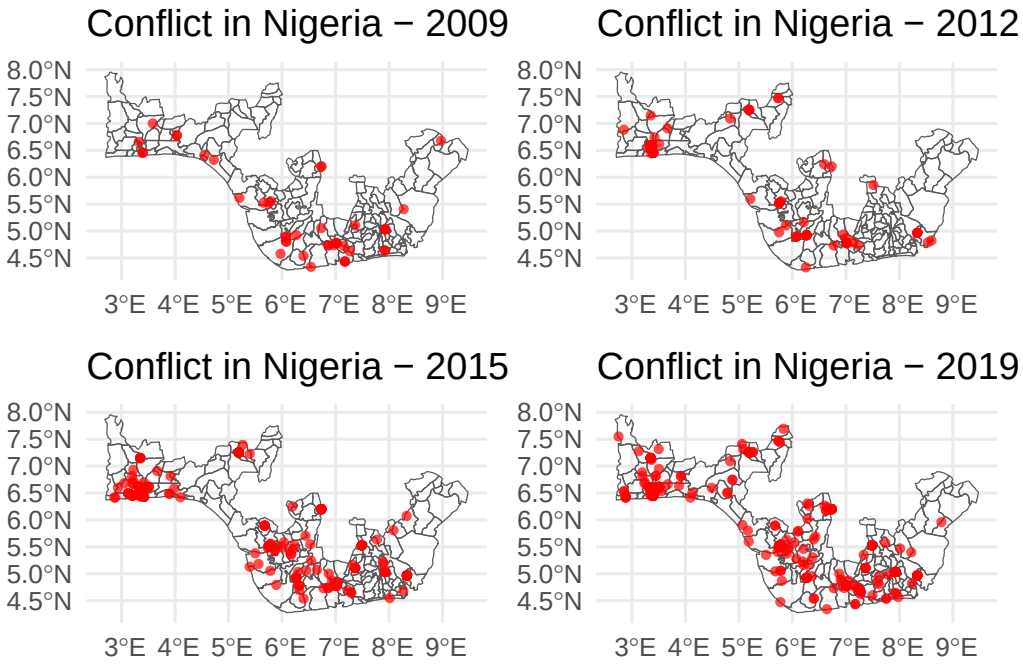


Figure 2.1 Nigeria conflict 2001- 2006- 2009- 2016

Figure 2.1 displays the spatial distribution of consumption-related conflict events across Nigerian coast in the years 2009, 2012, 2015 and 2019. The maps show a marked increase in conflicts between 2009 and 2019. In 2009, events were concentrated in a few clusters, while in 2019 they spread across most coastal regions, gradually extending westwards and inland. In 2015, there was a much more significant increase compared to 2012, and was greater than the gap between 2012 and 2009, suggesting an exponential growth dynamic.

Figure 2.2 shows the trend in the total number of conflict events per year (sum at ADM2 level). Focusing on the period 2009-2019, we can see that in the early years the number of incidents remained relatively low, but there has been a sharp increase since 2012, with progressive growth leading to a peak between 2015 and 2019. This trend confirms what has already been highlighted by the maps, namely a pattern of intensification and spread of conflicts over the decade.

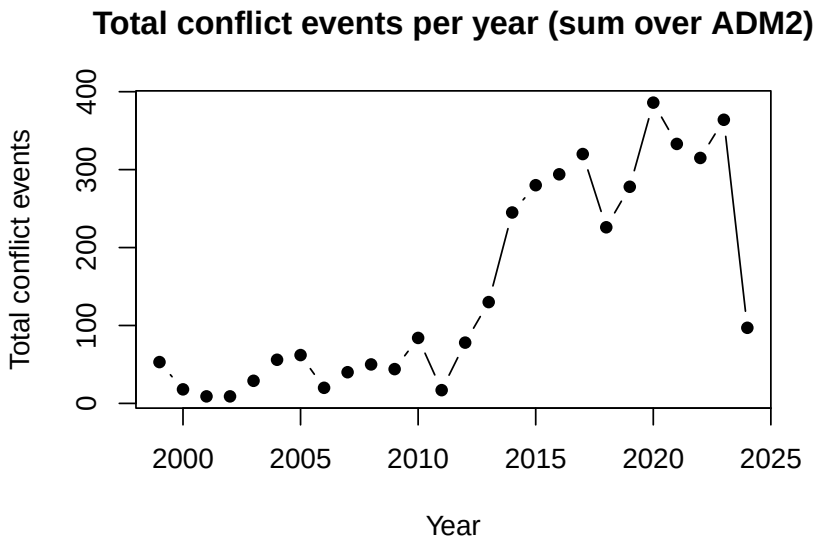


Figure 2.2 Conflict trend

In addition to the outcome variable, the empirical design requires the construction of an index of trade exposure. This measure combines the temporal variation of a national trade shock with the spatial heterogeneity of local exposure, ensuring that both time and geography contribute to identifying variation.
Index The index was constructed based on the intuition that a trade exposure shock can significantly influence the dynamics of conflicts. In particular, we consider how the impact of trade shocks varies geographically depending on the distance from trade ports: as highlighted by Berman and Couttenier (2015), regions closer to ports are in fact more exposed to external shocks. Following this intuition, we constructed an index that combines the measure of national connectivity with a local proxy for geographical accessibility to ports.

$$index = Shock \times LSCI \times 1/(distminnorm + \varepsilon) \tag{2.1}$$

Here, the shock is represented by Container Port Throughput, a dataset compiled by UNCTAD that estimates the total number of containers handled, by country, expressed in twenty-foot equivalent units; LSCI stands for Liner Shipping Connectivity, which measures a country's integration into global liner shipping networks, also from UNCTAD; finally, there is the subnational variable, which is the inverse of the normalized minimum distance to the nearest commercial port and ε is a very small positive constant introduced to avoid division by zero.

The main idea is that being close to commercial ports makes it easier to get to international markets. In other words, areas near ports are more exposed to international trade, while areas far away are relatively isolated. This insight is consistent with empirical evidence that regions closest to ports are most affected by global shocks, whether positive or negative, that are transmitted through trade (Berman

& Couttenier, 2015; McGuirk & Burke, 2020). Similar to the quasi-experimental approach by Amodio et al. (2021), who exploited Israel's 2008 ban on dual-use imports to the West Bank as an exogenous shock to trade, the present index captures how sudden or structural changes in trade integration may affect local economic and conflict dynamics. While their evidence shows that trade restrictions depress wages and increase violence, here the focus is to assess whether greater maritime accessibility and trade intensity can mitigate conflict in a developing-country context such as Nigeria.

The national component of the index is represented by the Liner Shipping Connectivity Index (LSCI), developed by UNCTAD (2025a) and frequently used in maritime trade literature as shown in Mishra et al. (2021) and Fugazza and Hoffmann (2017). The LSCI measures a country's structural capacity to integrate into global trade, based on variables such as the number of liner services, active operators, ships employed, ship size and total capacity in Twenty Foot Equivalent Units (TEU). It is internationally recognised as an indicator of trade accessibility and competitiveness, particularly relevant in developing countries where maritime transport is the main route for international trade. Empirical studies confirm that greater maritime connectivity reduces trade costs and promotes growth: for example, Fugazza and Hoffmann (2017) estimate that the absence of direct connections can reduce the value of bilateral exports, while the Review of Maritime Transport UNCTAD (2012) shows that improvements in the LSCI ranking are systematically associated with higher trade volumes and greater market access.

Foreign trade shock is measured using Container Port Throughput (CPT), a dataset compiled by UNCTAD (2025b) that reports the annual number of containers handled in each country's ports, expressed in TEU. CPT is used as an indicator of the intensity of maritime trade flows, as most international trade, in terms of volume, takes place by sea (Ayesu & Boateng, 2024). Meanwhile the LSCI captures the potential capacity of a national maritime transport network, the CPT reflects actual containerised traffic and thus provides a more direct measure of exposure to global supply and demand shocks. Empirical studies confirm that changes in CPT have tangible macroeconomic effects. For example, Ayesu and Boateng (2024) demonstrate that increased containerised traffic significantly boosts employment in African economies with maritime ports, highlighting its role as an engine of economic development. At the same time, the OECD/ITF report by Ducruet et al. (2014) points out that throughput also reveals structural vulnerabilities: in ports where efficiency is low, high levels of throughput can amplify the impact of external shocks due to congestion and longer periods of stay. From a policy perspective, UNCTAD UNCTAD (2012) recognises CPT as a key indicator of port development performance in emerging economies, highlighting its value as a standard measure of exposure to international trade. In this analysis, we consider the CPT as an exogenous source of variation in maritime trade flows which, interacting with measures of national connectivity (LSCI) and local port accessibility (distance from port), captures the heterogeneous transmission of global trade shocks to subnational regions.

Figure 2.3 presents the trend of the LSCI and CPT in Nigeria over the period 2009–2019. The two indicators show a similar trend: sustained growth until

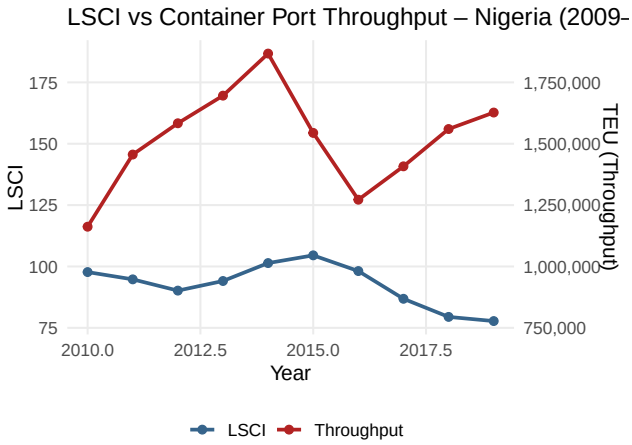
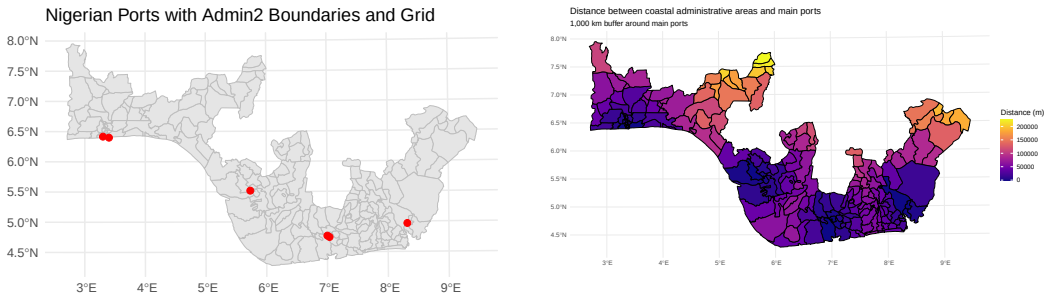


Figure 2.3 LSCI and CPT trend

2014–2015, followed by a contraction phase and a partial recovery in recent years. Their co-movement confirms that LSCI and CPT capture complementary dimensions of the same trade dynamic: the former reflects the structural capacity for integration into global trade, the latter the actual measure of the volume of containers handled. For this reason, both are included in the construction of the trade shock index used in the empirical analysis.

The role of distance from ports is crucial to understanding the heterogeneity of trade shocks. Following the intuition of gravity models Anderson and van Wincoop (2003), geographical distance remains a central determinant of trade costs and access to markets. In our framework, minimum distance from ports is therefore employed as a measure of local exposure to international trade shocks. (Anderson & van Wincoop, 2003) According to Berman and Couttenier (2015), regions located closer to major seaports are more directly integrated into international trade networks and therefore more strongly affected by external income shocks. Their findings show that positive shocks reduce the likelihood of conflict within these open regions, while the effect largely disappears in remote inland areas. This pattern suggests that better access to trade increases local incomes and raises the cost of using violence, thereby mitigating conflict. Following this idea, we incorporated distance from ports into the construction of my index. The logic is that Nigerian districts closer to seaports are more exposed to changes in global trade flows and should therefore exhibit a stronger response to consumption-related conflicts in the event of external shocks than districts located further inland. For each coastal ADM2, the distance from the nearest port was calculated, converted into kilometres and then normalised through a min–max transformation. To ensure comparability across units and to give greater weight to areas close to ports, the normalised distance was inverted so that higher values indicate stronger exposure to trade. Figure 2.4a shows the position of the six main ports overlapped for the ADM2 subnational regions. These ports correspond to those officially listed by the Nigerian Ports Authority and identified in the IMF PortWatch database as handling the largest shares of national maritime

trade activity. Starting from those ports the normalised minimum distance between each coastal administrative unit and the nearest commercial port was calculated. Figure 2.4b report the spatial distribution within a 1,000 km buffer. We can see that the coastal areas immediately adjacent to the ports, in dark purple, are naturally much more accessible, while the distance increases rapidly as one moves inland, highlighted by the lighter shades. This geographical heterogeneity forms the basis for the variable, which assigns greater exposure to areas closer to ports and progressively reduces that of more remote regions.



(a) Main trade ports **(b)** Buffer around main ports

Figure 2.4 Comparative maps: location of ports(a) and minimum distance from them(b)

While the LSCI index measures Nigeria's overall capacity to engage in global maritime trade, the minimum distance to ports is a reflection of regional disparities in the actual access to these connections. By integrating these two dimensions with the external shock represented by CPT, the index reflects both the macro structure of the port system and local differences in exposure. In this way, not only are general changes in the country's position within maritime networks captured, but also how such shocks are absorbed differently depending on geographical proximity to ports. The combination of these elements allows us to move from a uniform national perspective to a more detailed one, highlighting the territorial heterogeneity of the effects of maritime trade and their relevance to the dynamics of conflicts.

Nighlights Night-time lighting (NTL) data are widely used in the literature as an indicator of economic activity, especially in contexts where conventional economic statistics are scarce or unreliable. The underlying idea is that artificial night-time lighting is strongly correlated with income levels, infrastructure and access to energy, thus serving as a reliable indicator of local development (Amodio et al., 2024; Chen & Nordhaus, 2011; Henderson et al., 2012). In the case of Nigeria, night-time lights have been used to assess access to electricity and regional inequalities, confirming their strong association with patterns of socio-economic development. To construct this variable, we relied on the Harmonised Global Night-time Light Dataset (1992-2018) developed by Li et al. (2020), which combines the DMSP-OLS series with the latest VIIRS observations into a consistent time series. The harmonisation procedure corrects for satellite drift, blooming, and sensor inconsistencies, thus

ensuring temporal comparability. The dataset is available in yearly composites and can be aggregated spatially at multiple scales, making it particularly suitable for integration with ADM2-level units.For each coastal ADM2 and for each year, we calculated the logarithm of night-time light intensity to reduce asymmetry and facilitate interpretation in regressions. In addition, we defined a dummy variable equal to one for ADMs with values above the 75th percentile of the log-NTL distribution within the year. This distinction allows us to test whether trade shocks have differential effects between more and less economically developed regions.

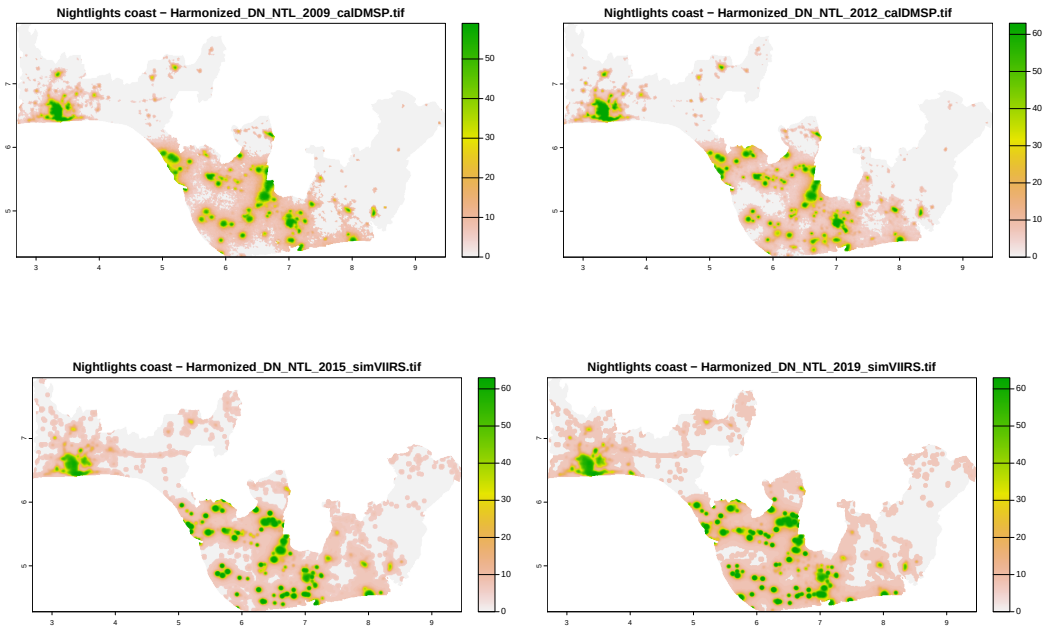


Figure 2.5 nightlights

Figure 2.5 shows the spatial distribution of night-time light intensity in Nigeria's coastal ADM2s. Light concentration is clearly clustered around Lagos, Port Harcourt, and other industrial centres, while rural areas show very limited brightness, consistent with the literature on energy poverty in Nigeria.

Moving on to temporal dynamics, Figure 2.6 tracks the average intensity of night-time lights over the period 2000–2018. The trend reveals an overall increase in luminosity, reflecting the gradual urbanisation of Nigeria, but also a significant decline around 2013 and 2016. These fluctuations are consistent with the technical issues documented in the harmonised dataset. In particular, Li et al. (2020) notes that the decline observed in the period 2015–2016 is also present in independent VIIRS composites and attributes it to sensor fluctuations and the transition from DMSP to VIIRS rather than an actual economic contraction. Similarly, Islam (2024) highlights the challenges posed by blooming, overglow, and inter-sensor calibration in the Nigerian context, emphasising that apparent discontinuities in time series may reflect data artifacts rather than real changes in development. It is important to note that such noise has been highlighted in previous studies as a limitation of

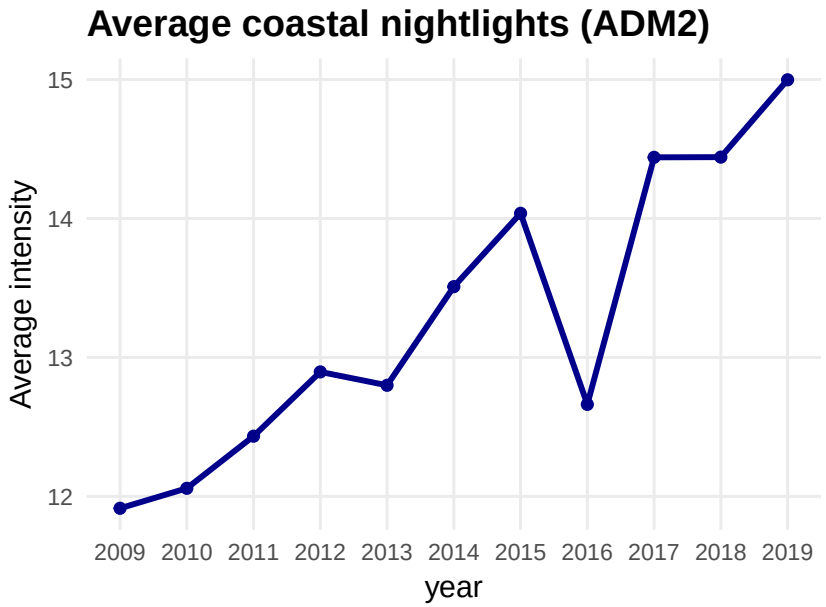


Figure 2.6 Average coastal nightlights

remote sensing for socio-economic monitoring, but the harmonisation procedure mitigates much of the distortion. We analyse the fluctuations in Appendix A
In summary, night-time lights provide a proxy for local economic activity. The expectation is that areas with lower night-time light intensity (i.e. poorer and less developed regions) should be more sensitive to foreign trade shocks, as consumption disruptions are more likely where households face tighter constraints on basic goods and services.

Geographical and sector characteristics. To account for spatial and structural differences across Nigeria's coastal economy, we classified ADM2 units and ports along two complementary dimensions. First, East–West regions were defined according to the median longitude of ADM2 centroids, distinguishing the eastern coastal belt—centered around Port Harcourt and Calabar—from the western belt dominated by Lagos and its surroundings. This partition captures historical and economic disparities in port activity and industrial development. Second, sectoral and trade-flow specializations were derived from the IMF PortWatch database, which provides information on the predominant categories of goods traded through each port. Following the data availability since 2019, ports were classified as either vegetable-oriented or chemical-oriented, the two main product groups after minerals, and as predominantly import- or export-oriented depending on the share of incoming and outgoing containerized flows. These distinctions are used later in the heterogeneity analysis to test whether the pacifying effect of trade exposure differs by regional position, sectoral composition, or trade direction. We report the sources for all these variables in Appendix B

Water availability We expect that regions with greater water availability are

those where economic activity is higher and more connected to national markets. Since the vegetable sector represents one of the main traded categories in Nigerian ports, second or third depending on the port, water resources are an essential input and thus a relevant proxy for agricultural income. For this reason, we use water availability as an indicator of agricultural activity and local economic potential (FAO & IIASA, 2020). The data come from FAO–GAEZ, which provides raster information on the mean quantity of water resources (or hydrological balance) available in each geographical unit. We calculate the mean value for every ADM2 area, obtaining a spatially continuous and relatively stable characteristic of the territory. This variable is then used to test how the pacifying effect of trade varies with the underlying resource environment.

Chapter 3

Methodoloy

We expect the economic impact of trade shocks to vary across different ADM2 units, with areas most exposed to international trade experiencing fewer consumption-related conflict events. Following this intuition, the index presented in the chapter Data was constructed, combining two national components that vary over time with a fixed spatial component. This approach is consistent with consolidated practices in the literature that combine aggregate shocks with sub-national heterogeneity (Autor et al., 2013).

The mechanism linking trade exposure and conflict is consistent with the opportunity-cost and grievance framework discussed by Amodio et al. (2021). Their evidence suggests that economic integration and access to trade can shape local conflict dynamics by altering individuals' economic incentives and perceptions of welfare. In this thesis, the focus is on the opposite mechanism: instead of trade restrictions, we study whether higher trade openness and proximity to maritime ports, which expand local economic opportunities, can mitigate the incidence of violence. The same channels highlighted by Amodio et al. (2021) are therefore expected to operate in reverse, with improved trade connectivity reducing both grievance and the opportunity to engage in conflict.

3.1 Conceptual framework

To better capture the mechanisms through which trade exposure affects social stability, we construct a specific subset of conflict events reflecting consumption- or income-related tensions rather than resource-seeking or political violence. Starting from the ACLED dataset (ACLED, 2025), we select events classified as protests or demostrations, the types less likely to have a link with resource predation. In particular, we identify income-related events through a keyword-based text filter, selecting those that mention terms associated with basic-needs or livelihood shocks, such as food, fuel, price, subsidy, market, ration, maize, rice, bread, or tax. Conflict incidence can be represented as (3.1) where Y_{it} captures opportunity-income effects (the effect of trade in income, reducing conflict) and R_{it} captures rent-seeking incentives (the opposite effect of trade that incentivises conflict).

$$Conflict_{it} = f(Y_{it}, R_{it}) \tag{3.1}$$

Our index captures exogenous variation in the income effect transmitted through trade shocks, holding the incentives largely constant by excluding rapacity-related events.

3.2 Identification Strategy

Identification relies on within district (ADM2) variation in exposure to national trade shocks over time. Because the trade shock (CPT × LSCI) is common across units while the distance is time-invariant, only the interaction between these components generates differential effects once ADM2 and year fixed effects are included. In our case, the CPT was delayed by one year to mitigate possible simultaneity issues, while the LSCI represents the degree of integration of the Nigerian port system into global maritime networks. Spatial heterogeneity is introduced through the minimum distance from ports, which serves as a proxy for accessibility to international markets. Following the intuition of (Berman & Couttenier, 2015), regions located closer to ports are more directly integrated into international trade networks and therefore more strongly affected by external shocks. In our context, this motivates the use of minimum distance from ports as a measure of local exposure, assuming that trade shocks are transmitted more intensely to areas with better access to maritime infrastructure. In this sense, the empirical design is consistent with the validity conditions discussed by (Goldsmith-Pinkham et al., 2020)
The empirical model also includes both spatial (ADM2) and temporal (year) fixed effects. In this way, given the same national shock, only differences in proximity to ports generate variation between units, allowing the heterogeneous effect of trade shocks on conflict to be isolated. To account for spatial heterogeneity the distances from ports where normalised. Since the relationship between trade exposure and conflict is expected to decrease with distance, the raw distances (measured in kilometers) were first inverted to capture proximity rather than remoteness. To ensure comparability across regions, the proximity measure was then rescaled between 0 and 1, so that values closer to 1 represent areas with higher accessibility to maritime trade. This transformation preserves relative spatial differences while avoiding scale distortions that could bias the exposure index.

3.3 Empirical Strategy

All models are estimated using OLS with two-way fixed effects (ADM2 and year). Standard errors are clustered at the ADM2 level to account for serial correlation within units over time. The empirical strategy consists of three steps. *First*, we estimate the basic model in two specifications: an aggregate version, in which the exposure index is considered in its overall form (Equation(3.2)), and a "split" version, which distinguishes administrative units based on the intensity of economic development measured by night-time lights (Equation (3.3)). This allows us to analyse both the average effect of the trade shock and the differences between more and less developed areas.

$$Conflict_{it} = \beta Index_{it} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.2}$$

$$Conflict_{it} = \beta_1 Index_{it}^{HigNL} + \beta_2 Index_{it}^{LowNL} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.3}$$

Second, we perform a falsification test to verify that the results do not simply reflect a spurious correlation between conflict and geographical proximity to port infrastructure. To this end, we reconstruct the index to include all ports in Nigeria, not just commercial ones. The idea is that if the effect were linked exclusively to physical proximity to a port, it should also emerge with this broader measure. Conversely, if the association is really driven by trade shocks, the results should only appear when considering ports that are actually relevant to international trade.
Third, beyond the baseline model, we explore whether the trade–conflict relationship varies across key structural dimensions. Following the idea that the effects of economic integration depend on pre-existing territorial and social characteristics ((Berman & Couttenier, 2015)), we analyse the heterogeneity of the effects across region, sector and environment. In particular, we distinguish between (i) eastern and western regions, (ii) the predominant type of industry traded, (iii) the predominant trade flow and (iv) water availability. These specifications allow us to investigate the differentiated transmission of trade shocks depending on the structural conditions of the territories. Equation 3.4 extends the baseline model to account for heterogeneity across structural or environmental dimensions. The interaction term captures whether the effect of trade exposure differs depending on regional location or sectoral specialization. A negative and statistically significant coefficient on β_2 would indicate that the pacifying effect of trade is stronger in the corresponding subgroup.

$$Conflict_{it} = \beta_1 Index_{it} + \beta_2 (Index_{it} \times H_i) + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.4}$$

Equation 3.4 account for water availability heterogeneity. $HighWater$ and $LowWater$ are dummies equal to one for ADM2 units above or below the median of the FAO–GAEZ water-availability distribution. In this analysis the coefficients $\beta 2$ and $\beta 3$ capture how the trade–conflict relationship differs across regions with different quantity of water resources

$$\begin{aligned} Conflict_{it} = {} & \beta_1 Index_{it} + \beta_2 (Index_{it} \times HighWater_i) + \beta_3 (Index_{it} \times LowWater_i) \\ & + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned} \tag{3.5}$$

Chapter 4

Results

In what follows, we present the results. Table 4.1 reports the estimates of the main coefficient from Equation (3.2) and (3.3).

Table 4.1 Correlation between port shocks and conflict: commercial vs all ports

	(1) Commercial: total	(2) Commercial: split	(3) All ports: total	(4) All ports: split
Total (commercial)				
indice2_main	-1.01e-07			
	(6.30e-08)			
Split (commercial)				
indice2_highNL		-1.12e-07		
		(7.80e-08)		
indice2_lowNL		-5.61e-08***		
		(5.43e-09)		
Total (all ports)				
indice2_total			-6.77e-08	
			(4.29e-08)	
Split (all ports)				
indice2_highNL				-9.32e-08
				(6.02e-08)
indice2_lowNL				-1.81e-08
				(1.82e-08)
log(nightlight)	-0.879***	-0.878***	-0.870***	-0.867***
	(0.249)	(0.248)	(0.249)	(0.249)
Observations	1,611	1,611	1,611	1,611
Adj. R^2	0.662	0.662	0.661	0.662
Within R^2	0.028	0.029	0.024	0.027
RMSE	1.806	1.805	1.810	1.807

Fixed effects: ADM2 and year. SE clustered at ADM2 level.
358 observations dropped due to RHS NA. Signif.: *** 0.01, ** 0.05, * 0.1.

It shows the results of the baseline specification, comparing the commercial ports with the broader set of Nigerian ports. Column (1) and (2) focus exclusively on the commercial ports, while Columns (3)–(4) include all ports as a falsification test. Starting from Column (1), the results indicates that the aggregate coefficient on the exposure index is small and not not statistically significant. However, when the sample is split between high- and low-nightlight areas, as in Column (2), a strong heterogeneity emerges. his split distinguishes ADM2 units by local economic development, proxied by night time lights (as detailed in the Data section): where the

areas with higher income are the ones with values above the 75 percentile (Amodio et al., 2024; Henderson et al., 2012).
In the areas with reatevely low levels of nightlights, the effect of the commercial shock is negative and statistically significant, suggesting that a greater exposure maritme trade reduces the likelihood of consumption-based conflicts In contrast the coefficient is close to zero in the more developed area. This model is coherent with the intuition that the external trade shocks have stronger effect where the local economies are less diversified and more vulnerable to changes in global market integration (Berman & Couttenier, 2015). Thus distance from commercial ports is particularly important in the less developed regions, amplifying the regulatory effect of the trade shocks. This finding is especially relevant given that less developed areas are also more likely to have conflict (Brückner & Ciccone, 2010; Collier & Hoeffler, 2004)

Columns (3)–(4) serve as a falsification test. If the relationship was determined solely by proximity with any port, not only the commercial one, we would expect similar effects when all ten Nigerian ports are included. Columns 3 and 4 extend the analysis, considering all nigerian ports, regardless their commercial activity. The results show weaker and not statistically significant coefficients, both for the aggregate and the splitted samples. The absence of an effect reinforces the interpretation that the mechanism is specificaly linked to international trade flows, rather than mere geographic proximity.

Overall, the results suggest that the link between trade and conflict is neither mechanic nor omnipresent, it rather depends commercial relevance of ports and the level of local economic development. The coefficients support the idea that international integration can provide stabilising benefit in the peripherical regions, and so the less developed ones economically speaking,while its marginal impact in more developed areas remains limited.

4.1 Heterogeneity

After establishing the baseline relationship, we explore whether the estimated effects vary across Nigeria's main macro-regions and according to local social fragmentation. Table 4.2 combines these dimensions of heterogeneity: Panel A shows how the trade–conflict link differs across regions and sectors, whereas Panel B reports the corresponding nightlight averages, illustrating the underlying economic disparities between groups.

Geography. Columns (1) investigates geographic heterogeneity: the index is interacted with East/West dummies defined by the median longitude of coastal ADM2 centroids (details in Appendix B.1). The estimated effect is more negative and statistically stronger in the eastern belt—areas that are poorer and more connected to maritime activity—consistent with a vulnerability-to-trade mechanism. A detailed discussion of the spatial correlation between the macro-regional partition,

Table 4.2 Regional and sectoral heterogeneity (combined)

Panel A. Index effects

	East vs West	Chemicals vs Vegetables	Export vs Import
indice2_est	-1.63e-07 *		
	(8.24e-08)		
indice2_west	-7.45e-09		
	(4.76e-08)		
indice2_main_veg		-1.63e-07 *	
		(8.24e-08)	
indice2_main_chem		-1.56e-07	
		(1.27e-07)	
indice2_main_exp			2.13e-09
			(3.29e-08)
indice2_main_imp			-2.55e-07 ***
			(5.84e-08)
Controls			
log(nightlight)	-0.884 ***	-0.884 ***	-0.884 ***
	(0.2493)	(0.2493)	(0.2470)
Observations 1,611	1,611	1,611	1,611
Adj. R^2	0.665	0.665	0.670
Within R^2	0.0361	0.0361	0.0499
RMSE	1.799	1.799	1.786

Panel B. Average nightlights (mean / median / sd)

Group	Mean NL	Median NL	SD NL	N
East	9.78	6.57	10.4	979
West	16.8	8.91	18.9	990
Chemicals	16.5	8.41	19.2	957
Vegetables	10.3	6.76	10.6	1,012
Export	16.4	9.17	17.9	1,199
Import	8.48	5.83	9.52	770

Fixed effects: ADM2 and year. SE clustered at ADM2 level. 358 obs. dropped due to RHS NA. Signif.: *** 0.01, ** 0.05, * 0.1. NL = nightlights (mean annual radiance, harmonized DMSP-OLS/VIIRS). ADM2-level averages, 2009–2019.

together with corresponding maps, is reported in Appendix B.1.

These patterns mirror the spatial distribution of economic development (Panel B). Nightlight data shows that western ADM2s exhibit higher and more dispersed nightlight values—largely driven by the Lagos metropolitan area—while eastern units are uniformly dimmer, reflecting a more rural and less industrialized economic structure. In this case, the gap between richer and poorer areas (as proxied by nightlights) is particularly pronounced. Detailed maps, descriptive statistics of night-lights by region, and additional robustness checks are reported in Appendix B.2.

Sector. Columns (2)-(3) reports the sector heterogeneity: the index is interacted with two different sets of dummies: (i) the predominant type of industry

traded through each port — either chemical or vegetable products (the two main sectors after minerals, based on data from IMF PortWatch, available since 2019 and used here as a proxy); and (ii) the predominant trade flow orientation, that is, whether the port handles mainly imports or exports (also based on data from IMF Portwatch). The estimated effect is negative and statistically significant for vegetables and imports. For presentation purposes, we refer to the coefficients as index_*veg* and index_*chem* (or index_*exp* and index_*imp*), where the former represents the baseline effect and the latter includes the interaction term. The stronger and statistically significant effects observed for vegetable and import-oriented ports suggest that not only proximity and structural economic conditions matter, but also the nature and direction of trade flows. The results are consistent with previous findings that trade in basic consumption goods, such as agricultural products, can mitigate conflict by stabilizing local prices and incomes ((McGuirk & Burke, 2020)). The effect for vegetable-oriented ports is also consistent with what is shown by Amodio et al. (2024): regions with lower nightlight intensity, typically linked to agricultural activity, are more vulnerable to economic shocks, suggesting that the welfare channel of trade is particularly relevant in such contexts. Similarly, import flows—particularly of essential goods—may enhance welfare and reduce consumption-related tensions (Brückner & Ciccone, 2010; Dube & Vargas, 2013). Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare-improving, rather than resource-based, channels.
Results are aligned with the underlying economic structure summarized in Panel B, where vegetable and import-oriented ports correspond to areas exhibiting more dispersed nightlight distributions, a pattern typically associated with more rural and less industrialized economies. Although the import/export distinction also reflects structural differences in local economies, import-oriented ports show a systematically stronger effect than all other categories. Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare-improving, rather than resource-based, channels.

We next investigate whether the pacifying effect of trade exposure varies with local resource endowments. Water availability, derived from the FAO–GAEZ database (FAO & IIASA, 2020), serves as a proxy for agricultural potential and economic activity, given its close link to crop productivity and the prominence of vegetables trade in Nigerian ports.

The intuition is that in ADM2 units with lower water availability, typically more fragile and rural, the trade index should reduce conflict by stabilizing consumption and providing alternative income channels. This mechanism is consistent with evidence that economic integration can mitigate consumption-related conflicts by improving access to markets and smoothing shocks (Amodio et al., 2024).

However, the empirical results in Table 4.3 reveal a more complex pattern. Contrary to expectations, the effect of trade exposure turns positive in areas with higher water availability—where agricultural and economic activities are already developed. This suggests that, rather than amplifying stability, trade may in these contexts reinforce existing inequalities or intensify competition over profitable

Table 4.3 Water Availability: Levels & Heterogeneity

	(1) year FE	(2) year FE	(3) ADM2+year FE	(4) ADM2+year FE
Index (overall / highNL / lowNL)	-3.59e-08	-3.95e-08 (highNL)	-1.327e-07*	-6.40e-08
	(1.21e-07)	(1.30e-07)	(6.09e-08)	(8.76e-08)
	–	-2.16e-08 (lowNL)	–	–
		(1.04e-07)		
Index × High water (median)	–	–	1.076e-07	–
			(6.88e-08)	
Index × Low water (median)	–	–	-1.40e-08	–
			(3.17e-08)	
Index × Water tercile: Medium	–	–	–	-2.597e-07**
				(8.75e-08)
Index × Water tercile: High	–	–	–	3.77e-08
				(9.38e-08)
Log Nightlights	0.902599**	0.902743**	-0.864138***	-0.868460***
	(0.199140)	(0.199352)	(0.245164)	(0.246453)
Water availability	-0.037963**	-0.037975**	–	–
	(0.010469)	(0.010490)		
Observations	1620	1620	1620	1620
Adj. R^2	0.102	0.101	0.664	0.668
Within R^2	0.0816	0.0816	0.0331	0.0454
RMSE	3.116	3.116	1.797	1.785

Note: (1)-(2) cluster year; (3)-(4) cluster ADM2. Baseline di gruppo/tercili omesse per collinearita'. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

*Metodo.*Water availability sono degli indicatori esterni di disponibilità idrica assegnati agli ADM2 e suddivisi per mediana o tercili. Le interazioni testano se la presenza di risorse idriche modula l'impatto.

markets (Berman et al., 2017; Michalopoulos & Papaioannou, 2013). Overall, the findings indicate that trade openness exerts a stabilizing influence mainly in less developed and resource-constrained local economies, a result in line with broader evidence on the heterogeneous development impacts of market integration (Amodio et al., 2024; Dix-Carneiro & Kovak, 2023).

Chapter 5

Conclusione

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Appendix

.1 conflict

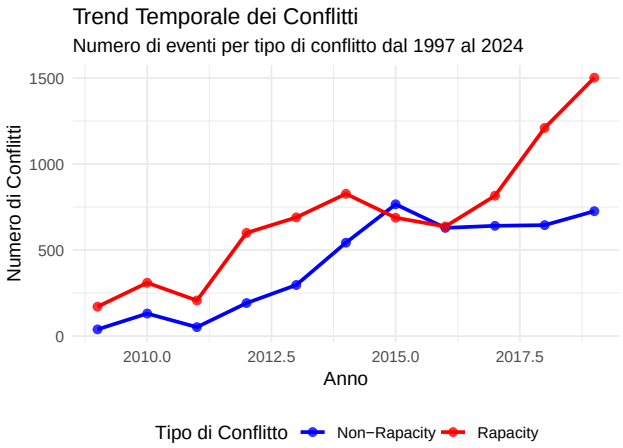


Figure .1 Descrizione dell'immagine.

.2 Nightlights

.3 Spatial heterogeneity

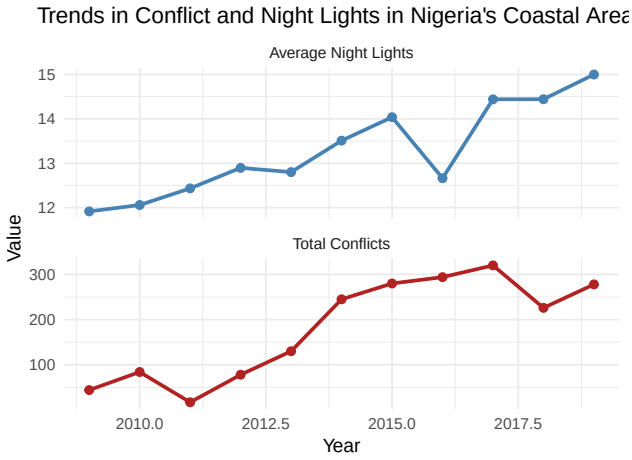


Figure .2 Descrizione dell'immagine.

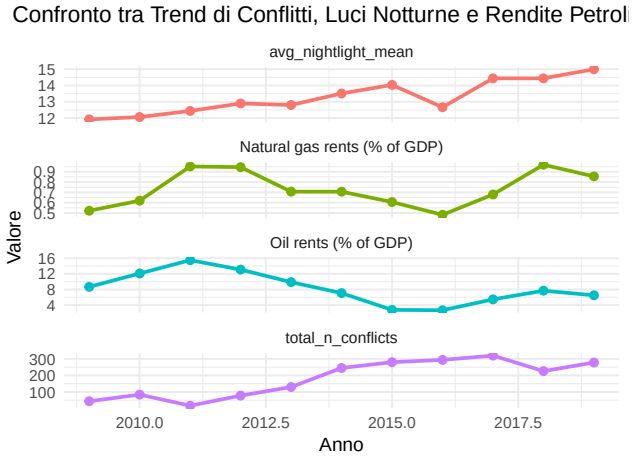


Figure .3 Descrizione dell'immagine.

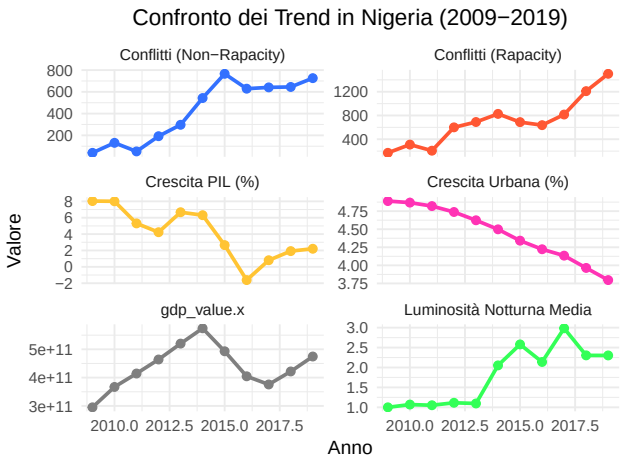


Figure .4 Descrizione dell'immagine.

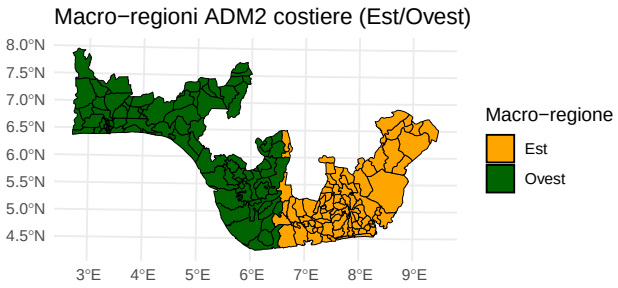


Figure .5 Descrizione dell'immagine.

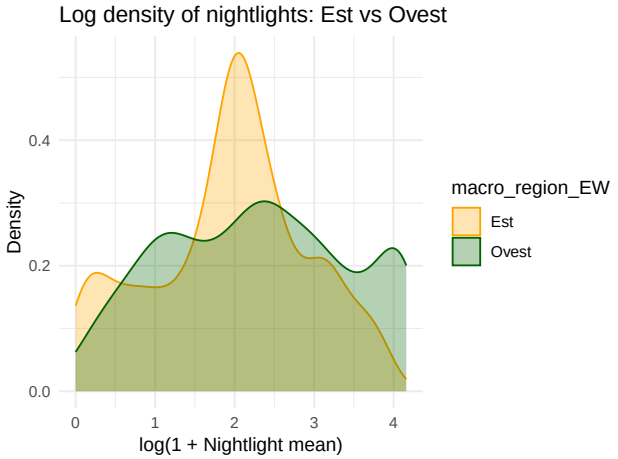


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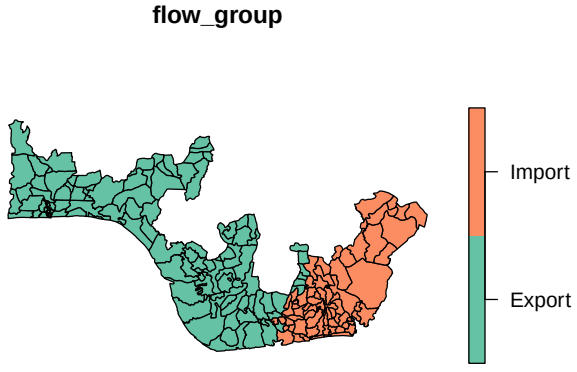


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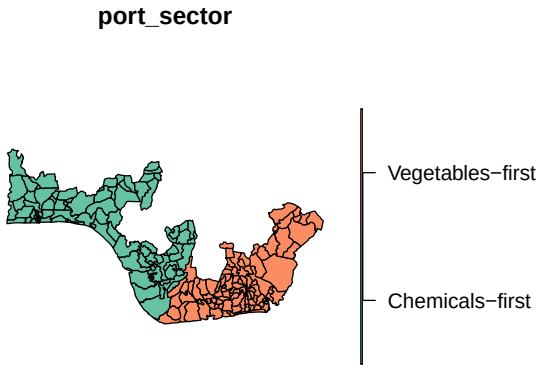


Figure .8 Descrizione dell'immagine.

Ringraziamenti

SCRIVI QUI I RINGRAZIAMENTI